

HAMMER ODDE FYR, Denmark

WMO: 061930

Lat: 55.2982N

Lon: 14.7738E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.6	-3.4	-8.9	1.8	-3.5	-7.3	2.0	-2.6	20.3	3.4	18.4	3.2	5.0	120	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	23.6	19.1	22.4	18.7	21.2	18.1	20.2	22.6	19.3	21.6	18.5	20.7	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	13.9	21.7	18.3	13.2	20.8	17.6	12.6	20.0	57.9	22.8	55.1	21.7	52.4	20.7	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.2	13.7	DB	-6.1	27.3	1.9	2.5	-7.4	29.0	-8.5	30.5	-9.6	31.9	-11.0	33.7	(4)
(5)			WB	-7.1	21.5	1.7	1.8	-8.3	22.8	-9.4	23.8	-10.3	24.8	-11.6	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	1.8	1.6	2.9	6.1	10.4	14.6	17.6	18.0	14.9	10.6	6.6	3.6
	DBStd	6.38	2.77	2.76	2.52	2.60	2.73	2.29	2.35	2.11	2.14	2.57	2.51	2.82
	HDD10.0	1186	254	236	221	121	28	1	0	0	0	23	103	198
	HDD18.3	3417	513	469	479	366	247	116	43	32	106	239	351	457
	CDD10.0	856	0	0	0	4	40	137	236	247	147	43	2	0
	CDD18.3	46	0	0	0	0	0	2	20	21	2	0	0	0
	CDH23.3	65	0	0	0	1	1	6	30	26	2	0	0	0
	CDH26.7	8	0	0	0	0	0	1	4	4	0	0	0	0
Wind	WSAvg	6.5	7.7	7.3	6.7	5.5	5.3	5.8	5.6	5.9	6.7	6.8	7.1	7.8
Precipitation	PrecAvg	538	46	29	34	33	33	41	49	49	57	51	64	55
	PrecMax	691	101	88	66	96	106	89	132	177	121	179	107	113
	PrecMin	377	5	6	4	3	4	7	5	2	6	5	19	13
	PrecStd	78	20	17	18	24	21	22	32	40	28	38	27	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	9.5	11.5	17.7	20.2	23.9	26.0	26.2	23.0	18.1	13.1	9.9
		MCWB	7.2	7.1	7.7	10.9	14.5	17.7	20.6	20.1	17.9	13.8	11.4	8.7
	2%	DB	7.0	7.2	8.5	13.7	17.6	21.0	23.7	23.4	20.5	16.0	11.7	8.7
		MCWB	6.0	5.6	6.4	9.7	13.7	17.0	19.6	19.3	17.0	13.9	10.6	7.8
	5%	DB	6.1	5.8	7.1	11.3	15.7	19.4	22.5	22.2	19.0	14.9	10.7	7.9
		MCWB	5.3	4.7	5.5	8.4	12.6	16.0	19.1	18.9	16.2	13.2	9.6	7.1
	10%	DB	5.3	5.0	6.0	9.7	14.3	18.1	21.3	21.2	17.8	14.0	9.9	7.1
		MCWB	4.6	4.2	4.7	7.5	11.7	15.3	18.4	18.2	15.5	12.6	8.8	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.3	8.3	12.0	16.0	18.9	22.2	21.1	19.0	15.2	12.0	9.2
		MCDB	8.2	8.8	10.2	14.8	18.8	22.1	24.0	24.2	21.3	16.5	12.5	9.5
	2%	WB	6.2	5.9	6.7	10.1	14.3	17.5	20.5	20.2	17.5	14.2	10.8	8.1
		MCDB	6.8	6.8	8.0	13.0	16.7	20.1	22.8	22.6	19.5	15.5	11.5	8.6
	5%	WB	5.4	4.9	5.7	8.8	13.1	16.5	19.5	19.4	16.6	13.5	9.8	7.2
		MCDB	6.0	5.6	6.8	11.0	15.2	18.7	21.9	21.6	18.3	14.7	10.6	7.8
	10%	WB	4.7	4.2	4.8	7.6	12.0	15.6	18.7	18.6	15.8	12.7	8.9	6.4
		MCDB	5.2	4.9	5.7	9.3	13.9	17.7	20.9	20.7	17.4	13.8	9.7	7.0
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.6	3.2	4.4	4.8	4.6	4.4	4.1	3.3	2.9	2.5	2.5
		MCDBR	2.7	3.6	5.2	7.6	7.5	6.7	6.3	5.6	5.1	4.0	3.1	2.6
		MCWBR	2.6	2.7	3.7	4.6	4.6	4.2	3.7	3.6	3.4	2.9	2.7	2.7
	5% WB	MCDBR	2.6	3.3	4.8	7.0	6.9	6.0	5.7	4.9	4.3	3.6	3.0	2.7
		MCWBR	2.7	2.7	3.6	4.6	4.7	4.3	3.6	3.5	3.3	2.9	2.8	2.8
Clear-Sky Solar Irradiance	taub	0.321	0.344	0.367	0.400	0.397	0.389	0.413	0.415	0.381	0.365	0.346	0.314	
	taud	2.299	2.316	2.303	2.279	2.355	2.408	2.360	2.356	2.424	2.386	2.322	2.287	
	Ebn at Noon	596	722	796	820	846	857	826	798	778	690	556	511	
	Edh at Noon	56	78	99	116	114	109	113	107	88	72	55	46	
All-Sky Solar Radiation	RadAvg	0.42	1.04	2.45	4.33	5.69	6.11	5.56	4.48	3.16	1.50	0.58	0.29	
	RadStd	0.08	0.19	0.23	0.42	0.55	0.37	0.60	0.33	0.31	0.24	0.10	0.04	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-102	N/A	N/A	(46)
(47)	Regional (51 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-154	+60	N/A	(47)

Nomenclature: See separate page